SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 40%

QUESTIONS: 4 **Total Marks:20**

Annexure A
SHORT ANSWER TEST

Code of Conduct:

*I **Manoj A / 192421295** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Student Name: Manoj A
Register No.: 192421295
Course: CSA16 – Data Warehousing and Data Mining

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Annexure A

Short Answer Test

1.Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.

(a) Enumerate three classes of schemas that are popularly used for modeling data warehouses.

(b) Draw a schema diagram for the above data warehouse using one of the schema.

(c) To obtain the same list, write an SQL query assuming the data is stored in a relational database with the schema fee (day, month, year, doctor, hospital, patient, count, charge).

(5 Marks)

2.Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester, and instructor, and two measures count and avg grade. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the avg grade measure stores the actual course grade of the student. At higher conceptual levels, avg grade stores the average grade for the given combination. (5 Marks)

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [student,course, semester,instructor], what specific OLAP operations

(e.g., roll-up from semester to year) should one perform in order to list the average grade of CS courses

for each Big University student.

(c) If each dimension has five levels (including all), such as "student < major < status < university < all", how many cuboids will this cube contain (including the base and apex cuboids)?

3.Design a data warehouse for a regional weather bureau. The weather bureau has about 1,000 probes, which are scattered throughout various land and ocean locations in the region to collect basic weather data, including air pressure, temperature, and precipitation at each hour. All data are sent to the central station, which has collected such data for over 10 years. Your design should facilitate efficient querying and on-line analytical processing, and derive general weather patterns in multidimensional space. (5 Marks)

4. A data cube, C, has n dimensions, and each dimension has exactly p distinct values in the base cuboid. (5 Marks)

Assume that there are no concept hierarchies associated with the dimensions.

(a) What is the maximum number of cells possible in the base cuboid?

(b) What is the minimum number of cells possible in the base cuboid?

(c) What is the maximum number of cells possible (including both base cells and aggregate cells) in the

data cube, C?

(d) What is the minimum number of cells possible in the data cube, C?

ANSWERS

Name: Manoj A Register No.: 192421295

CSA16 – Data Warehousing and Data Mining | Assessment Tool 1 (CO1)

Question 1 (5 Marks)

(a) Three classes of schemas used for modelling data warehouses

- **Star schema** – one central fact table holding the measures and foreign keys, connected directly to one de-normalised table per dimension (no further normalisation of the dimension tables).
- **Snowflake schema** – a variant of the star schema in which the dimension tables are normalised into several related tables (one per level of a concept hierarchy), giving a snowflake-like shape.
- **Fact constellation (galaxy) schema** – several fact tables share dimension tables (e.g. a "visit" fact table and a "billing" fact table sharing the time, doctor and patient dimensions).

(b) Schema diagram (star schema)

Dimensions: **time**, **doctor**, **patient**. Measures: **count** (number of visits) and **charge** (fee charged by the doctor to the patient). The fact table's primary key is the combination of the three foreign keys.

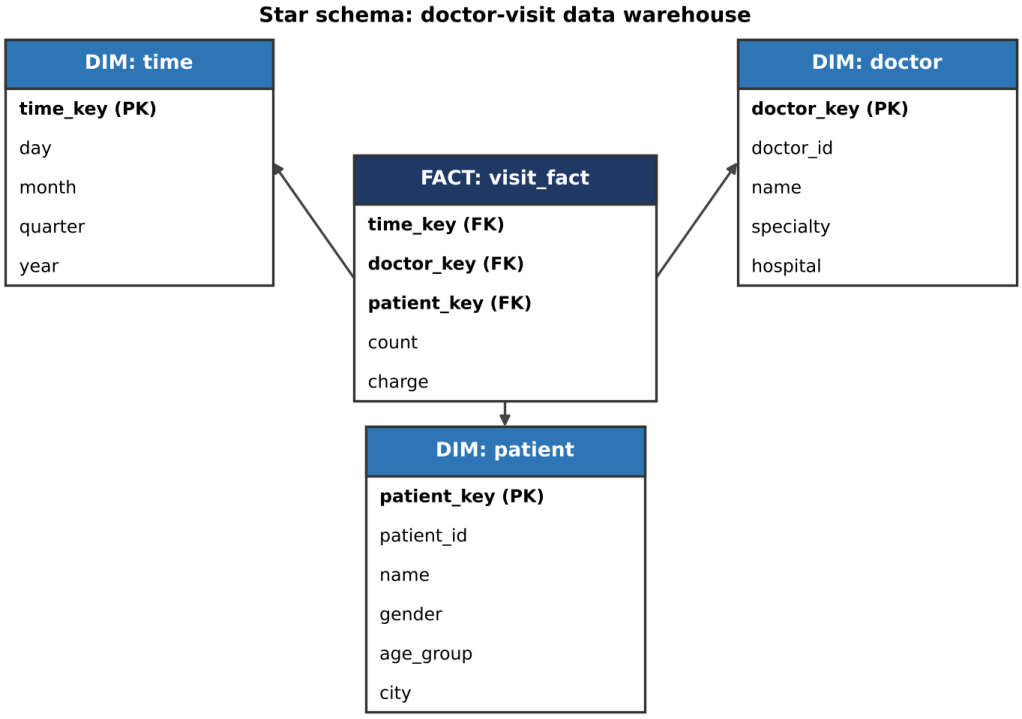


Figure 1: Star schema for the doctor–patient visit data warehouse

(c) SQL query

The question refers to "the same list" as the OLAP result, which (in the standard form of this exercise) is the **total fee collected by each doctor in a given year**. Taking the year 2010 as the example, the OLAP steps on the base cuboid [day, doctor, patient] are: roll-up time from

day to year, roll-up patient to all, and slice on year = 2010. The equivalent SQL on the relational table fee(day, month, year, doctor, hospital, patient, count, charge) is:

```
SELECT   doctor,
         SUM(charge) AS total_fee,
         SUM(count)  AS total_visits
FROM     fee
WHERE    year = 2010
GROUP BY doctor
ORDER BY doctor;
```

To list fees per doctor per hospital, add hospital to the SELECT and GROUP BY lists; to change the year, change the WHERE condition.

Question 2 (5 Marks)

(a) Snowflake schema diagram

Fact table **grade_fact** has the foreign keys of the four dimensions and the measures **count** and **avg_grade**. Each dimension is normalised along its concept hierarchy (student < major < status < university; course < department; semester < year; instructor < department).

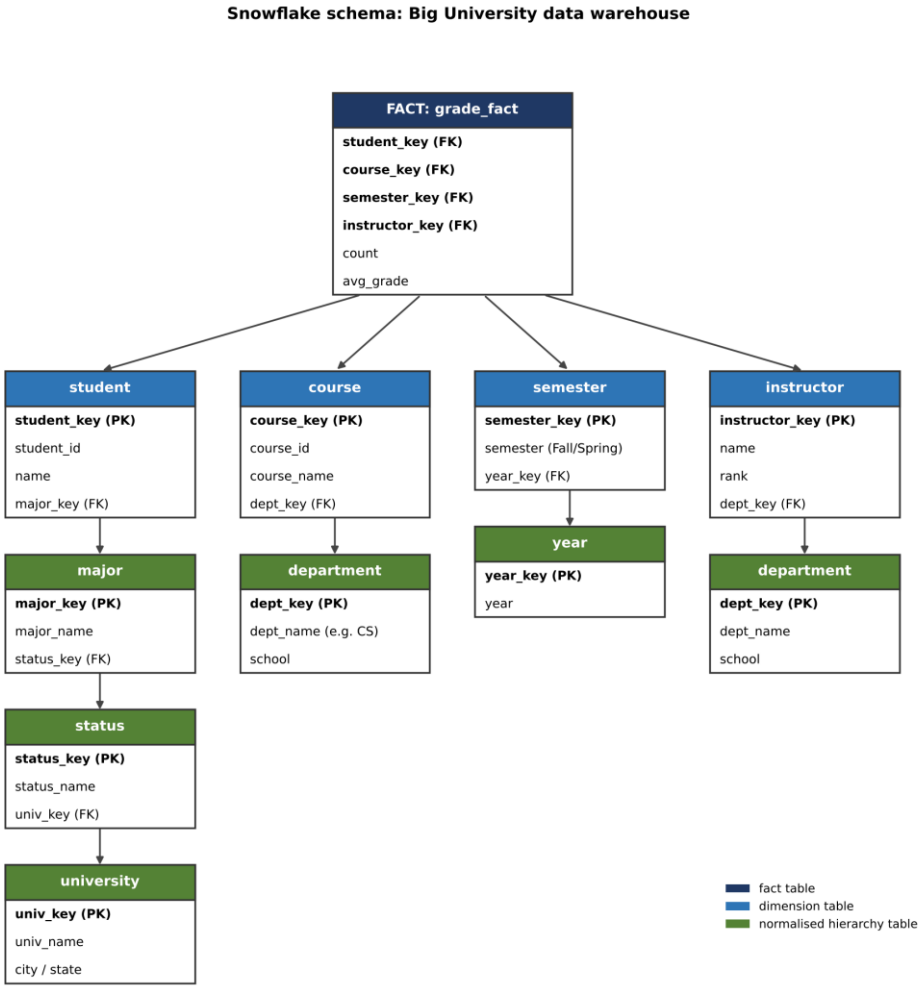


Figure 2: Snowflake schema for the Big University data warehouse

(b) OLAP operations to list the average grade of CS courses for each student

Start from the base cuboid [student, course, semester, instructor] and perform:

- **Roll-up on course** from course to department (so the course dimension is at department level).
- **Slice (selection) on department = "CS"** – keeps only the CS courses.
- **Roll-up on semester** from semester to all (ignore the term; one value over all terms).
- **Roll-up on instructor** from instructor to all (ignore who taught the course).
- **Keep student at its lowest level** (no roll-up on student), so the result has one row per student.

The result is the sub-cuboid [student, department = CS, all, all], giving one avg_grade per Big University student. Because avg_grade is an average, each roll-up must compute a **weighted average**: Σ(count × avg_grade) / Σ(count), not a plain average of averages.

(c) Number of cuboids

A cuboid is obtained by choosing one level (out of 5, including "all") for each of the 4 dimensions independently, so

$$\text{Number of cuboids} = 5 \times 5 \times 5 \times 5 = 5^4 = \mathbf{625}$$

This includes the base cuboid [student, course, semester, instructor] (lowest level in every dimension) and the apex cuboid [all, all, all, all]. In general, with n dimensions and L_i levels in dimension i, there are $\Pi\, L_i$ cuboids.

Question 3 (5 Marks): Data warehouse for a regional weather bureau

1. Requirements and assumptions

- About **1,000 probes** (land and ocean) each report **air pressure, temperature and precipitation every hour**; data are sent to a central station and kept for **10+ years**.
- Goals: fast ad-hoc queries and OLAP (roll-up, drill-down, slice, dice) and discovery of general weather patterns in multidimensional space.
- Assumed for sizing: about 50 districts grouped into 5 regions.

2. Schema design

A star schema (with the location hierarchy snowflaked) is used. Grain of the fact table = **one probe, one hour**. Dimensions: **Time** and **Probe/Location**; measures: air pressure, temperature, precipitation. Averages are stored as sum and count (algebraic measures) so they can be rolled up correctly; min and max are distributive and also stored.

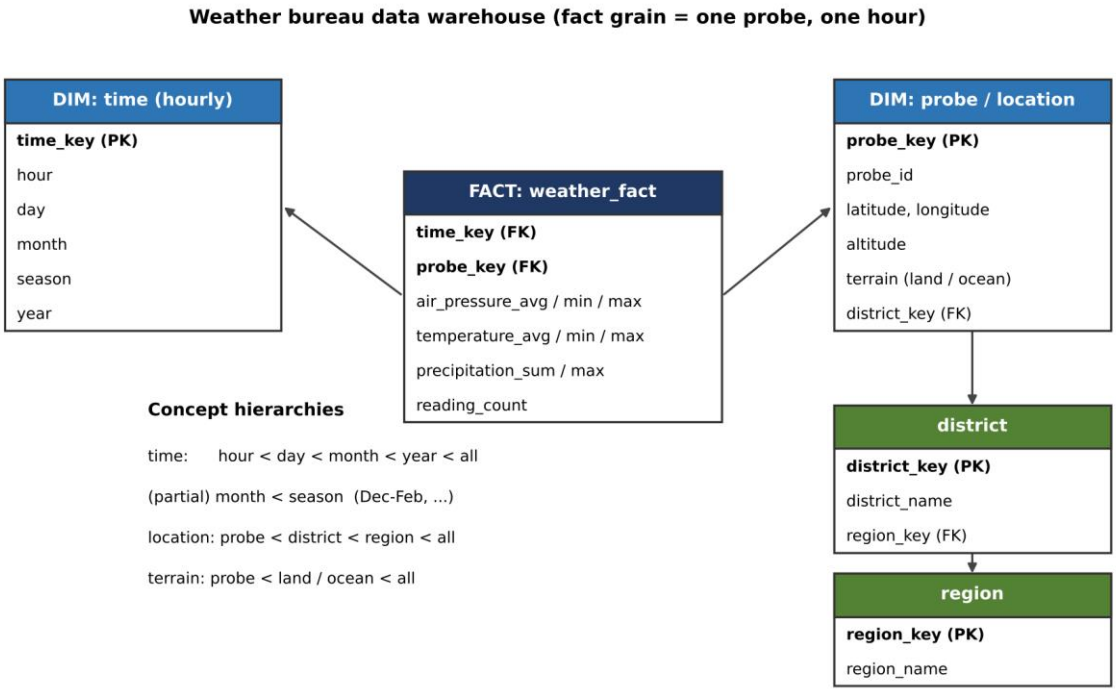


Figure 3: Weather data warehouse schema and concept hierarchies

3. Dimensions, hierarchies and measures

Item	Design
Time dimension	hour < day < month < year < all; also month < season (partial order), day-of-week, holiday flag
Probe / Location dimension	probe < district < region < all; attributes: latitude, longitude, altitude, terrain (land / ocean) – allows slicing by land vs ocean
Measures	air_pressure (avg, min, max), temperature (avg, min, max), precipitation (sum, max), reading_count
Optional dimension	weather condition (e.g. clear, rain, storm) derived from the readings, for pattern analysis

4. Size of the base cuboid

1,000 probes × 24 hours × 365 days × 10 years = **87.6 million** fact rows (≈ 3.5 GB at about 40 bytes per row). Every probe reports every hour, so the cube is dense, and a column-oriented or compressed fact table suits it well.

5. Data cube and materialisation

With the hierarchies above there are 5 time levels × 4 location levels = **20 cuboids**. Full materialisation is unnecessary; the base cuboid plus a few small, frequently used cuboids are precomputed and the rest are derived from the smallest materialised ancestor:

Cuboid (time, location)	Cells	Materialise?
(hour, probe) – base	87,600,000	Yes (base)
(day, probe)	3,650,000	Yes
(day, district)	182,500	Yes
(month, district)	6,000	Yes
(month, region)	600	Yes
(year, region)	50	Yes
(all, all) – apex	1	Yes

6. Supporting efficient querying and OLAP

- **Partition** the fact table by year/month so queries touch only the needed partitions; keep older years on cheaper storage.
- **Bitmap indexes** on low-cardinality attributes (terrain, region, season, month) and B-tree/clustered index on (time_key, probe_key).
- **Precomputed aggregates** (table above) answer most roll-up queries directly; ROLAP/MOLAP engine for drill-down to hourly data.
- **ETL**: hourly feed from the central station → validate (range checks, sensor faults), fill missing readings with NULL, standardise units, load incrementally; dimension tables loaded first.

7. Deriving general weather patterns

- **Roll-up / drill-down:** average temperature by region and month, then drill down to days of an unusual month.
- **Slice / dice:** ocean probes only; winter months in the northern region.
- **Pivot** (region × season) tables for comparing years; **trend** analysis of yearly averages (climate change signals).
- **Data mining on the cube:** clustering of probes with similar climate, association of pressure drop with heavy rainfall, outlier (exception) detection for storms, classification/prediction of rain from pressure and temperature.

Question 4 (5 Marks)

Cube C has n dimensions, each with exactly p distinct values in the base cuboid and no concept hierarchies, so every dimension has only two levels: its p values and "all" (*).

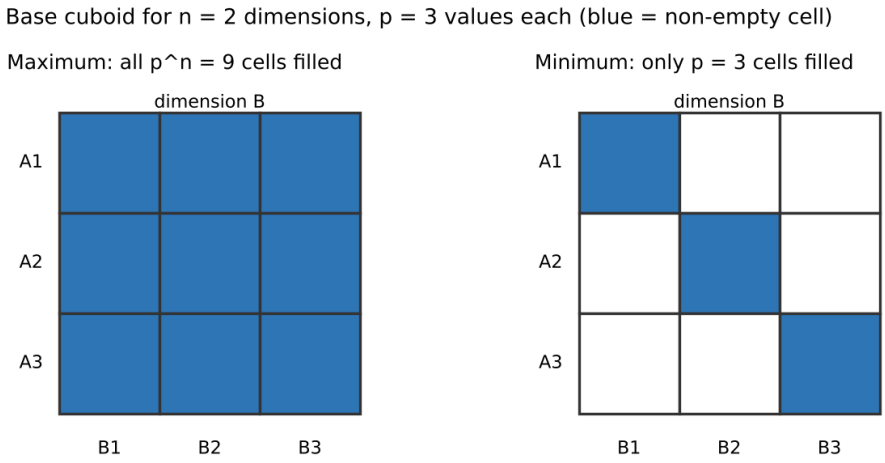


Figure 4: Maximum and minimum number of non-empty base cells (n = 2, p = 3)

Part	Answer	Reason
(a) Maximum cells in base cuboid	$\mathbf{p^n}$	Each of the n dimensions independently takes one of p values, giving $p \times p \times \ldots \times p = p^n$ combinations; every combination can be non-empty.
(b) Minimum cells in base cuboid	**p**	Each dimension must show all p of its values, so at least p cells are needed. p cells suffice, e.g. the "diagonal" $(v_1,v_1,\ldots,v_1)$, $(v_2,v_2,\ldots,v_2)$, …, $(v_p,v_p,\ldots,v_p)$.
(c) Maximum cells in the whole cube (base + aggregate)	$\mathbf{(p + 1)^n}$	In each dimension a cell takes one of p values or "*". That is p + 1 choices per dimension, so $(p + 1)^n$ cells in all cuboids together (reached when the base cuboid is full).
(d) Minimum cells in the whole cube	$\mathbf{(2^n - 1)\cdot p + 1}$	Take the diagonal base cells of (b). Each of the $2^n - 1$ non-apex cuboids (any non-empty subset of kept dimensions) still has exactly p distinct cells, and the apex cuboid has 1 cell: $(2^n - 1)p + 1$.

Check (n = 2, p = 3): maximum base = 9, minimum base = 3, maximum cube = $4^2 = 16$, minimum cube = $3\cdot3 + 1 = 10$. (Verified by brute-force enumeration of all valid base cuboids for small n and p.)